02 — Functional Requirements

Item	Details
Course	DBMS Project — Semester 5
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Stack	React + Vite; Node.js + Express; MySQL; MongoDB
Repository	https://github.com/snehajisho/Satellite_Data_Provenance_Earth_Observatory-DBMS_Project

4. Functional Requirements

4.1 Functional requirement catalogue

ID	Requirement	Description
FR-01	Registration	Register with username, email and password; required fields are validated and password length must be at least eight characters.
FR-02	Password storage	Hash passwords with bcrypt before inserting them into USERS.
FR-03	Login	Validate email/password and issue a signed JWT for valid credentials.
FR-04	Session handling	Frontend stores token/user data in local storage and attaches a bearer token to API requests.
FR-05	Authorization	Distinguish USER and ADMIN; administrator-only routes reject non-admin users.
FR-06	Health check	Expose /api/health to check API and MySQL connectivity.
FR-07	Satellite retrieval	Authenticated users can retrieve satellite records through the satellite API.
FR-08	Observation retrieval	Authenticated users can retrieve all observations and an observation by ID.
FR-09	Observation write controls	Observation creation is rejected; update/delete operations require authentication and administrator authorization.
FR-10	Metadata retrieval	Retrieve all MongoDB observation metadata and metadata by observation ID.
FR-11	Location search	Search metadata by location name using a case-insensitive regular-expression match.

ID	Requirement	Description
FR-12	Satellite metadata filter	Retrieve metadata documents for a specified satellite ID.
FR-13	Provenance	Authenticated users can retrieve processing activities associated with a data product, including input/output product information.
FR-14	Image delivery	Serve image assets from the database/mongodb/images directory through the Express /images static route.
FR-15	Frontend navigation	Provide Home, Satellites, Observations, Analytics, Provenance and Quality views.
FR-16	Location exploration	Allow users to search metadata by geographic location from the home/quality experience.
FR-17	Data presentation	Display satellite, observation, metadata, quality and processing context returned by the APIs.

4.2 Non-functional considerations

- **Maintainability:** frontend, backend routes and database scripts are separated into dedicated project directories.
- **Integrity:** relational constraints and triggers enforce important data rules.
- **Security:** bcrypt, JWT verification, role checks and parameterized SQL statements are used in backend access paths.
- **Interoperability:** React communicates with Express through HTTP/JSON APIs.
- **Flexibility:** MongoDB stores nested metadata documents.

4.3 Authorization summary

Exact permissions are enforced by the backend route middleware. The table summarizes the implemented authorization model; endpoint-specific route code remains authoritative.

Operation	USER	ADMIN
Register/login	Allowed	Allowed
Protected read endpoints	Allowed when authenticated	Allowed when authenticated
Administrator-only updates/deletes	Denied	Allowed where route permits
Create observation	Denied	Denied